SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA0307	**Course Title:**	Natural Language Processing
CO Assessed:	CO5 – Articulation	**Assessment:**	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	AI & ML / 2024	**Date:**	13/08/2026

Code of Conduct

I _D. Sri Anjaneya / 192425219_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _D. Sri Anjaneya_

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A – Constraint-Based Problem Solving

1. **Given the following paragraph:**
   "John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

   Tasks: a) Identify all referring expressions (mentions) and possible antecedents.

   b) Apply the following **constraints** to resolve the references:

   - Gender and number agreement
   - Recency (prefer recent antecedents)
   - Semantic compatibility (e.g., animacy and action suitability) • Coherence (maintain consistent entity chains)

   c) Draw a **constraint graph** or table showing variables, domains, and which constraints eliminate possible links.

   d) Provide the final resolved coreference chains and rewrite the paragraph with clear references.

   e) Justify the priority order of your constraints and discuss what would happen if the recency constraint is relaxed.

2. **Conversation History:** User: "I have an important exam tomorrow but I'm not able to concentrate."

   Required Dialog Act: Advise + Encourage (Give advice and motivate the student).

   Constraints:

1. Maintain **entity coherence** (reuse "exam", "concentrate", "you").
2. Use **discourse relations** (Cause-Effect or Elaboration).
3. Include at least **two** of these keywords: focus, break, confident.
4. Response length: 2–3 sentences only.
5. Logical consistency (no contradictory statements).
6. Politeness and positive tone.

   Tasks: a) Identify the relevant discourse planning steps needed before generation.

   b) Generate **three** different possible responses that satisfy **all** the above constraints.

   c) Evaluate the three responses and select the best one. Justify your choice using coherence and constraint satisfaction criteria.

   d) Explain how violating any two constraints would affect the quality of the dialog.

3. **Source Sentence:** "The bank by the river flooded after the storm, but it was saved by quick action."

   Note: "bank" is ambiguous (financial institution or riverbank).

Constraints:

1. Correct **Word Sense Disambiguation** using context.

2. Create a **predicate logic** representation (e.g., using predicates like flood(x), location(x,y)).
3. Maintain **discourse structure** (Contrast relation between the two clauses).
4. Preserve all important entities and relations (no information loss).
5. Ensure **text coherence** in the generated output.

Tasks: a) Resolve the ambiguous word “bank” and justify using constraints.

b) Write a predicate logic representation of the sentence.

c) Generate a good **target language sentence** (in any Indian language or simple English paraphrase) that satisfies all constraints.

d) Draw a simple **discourse tree** (RST-style) showing the rhetorical relation.

e) Discuss the advantages of using constraint-based approach over pure statistical machine translation for this sentence

1. Coreference Resolution Using Constraint-Based Problem Solving

Given Paragraph

"John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

a) Referring Expressions and Possible Antecedents

A **referring expression (mention)** is a word or phrase that refers to an entity mentioned earlier or understood from the context.

Referring Expression	Possible Antecedents	Initial Possibilities
He	John, Mary	John, Mary
She	John, Mary	John, Mary
it	ball, park	ball, park
him	John, Mary, dog	John, Mary, dog
they	John, Mary, dog	John, Mary, dog
all	John, Mary, dog	John, Mary, dog

The noun phrases **John, Mary, park, ball, and dog** are the main explicitly introduced entities.

b) Applying the Constraints

Constraint 1: Gender and Number Agreement

This constraint eliminates antecedents that do not agree grammatically with the referring expression.

- **He** → masculine singular → **John**
- **She** → feminine singular → **Mary**
- **it** → singular, non-human object → **ball** is the strongest candidate
- **him** → masculine singular → **John or dog**
- **they** → plural → **John + Mary + dog**
- **all** → plural group → **John + Mary + dog**

Therefore:

- He → John
- She → Mary
- it → ball
- him → John/dog
- they → John + Mary + dog

Constraint 2: Recency

Recency prefers the most recently mentioned compatible entity.

For **him**, the immediately preceding entities include:

"She wanted to play with it. **The dog** chased him…"

The dog is very recent, and grammatically **him** could refer to a male human or an animal. Therefore, recency initially favors **dog**.

However, recency cannot be applied alone because semantic compatibility and discourse coherence must also be considered.

Constraint 3: Semantic Compatibility

Semantic compatibility checks whether the proposed antecedent makes sense with the action.

The sentence is:

"The dog chased him excitedly."

A dog can naturally chase a person or another animal. Both **John** and **Mary** are possible objects of chasing, but **him** cannot refer to Mary because of gender agreement.

Thus:

- him → John or dog grammatically/semantically
- The interpretation **dog chased dog** is possible in isolation but is less natural in this context.
- **dog chased John** is semantically natural and creates a coherent event.

For **it**:

"She wanted to play with it."

A **ball** is strongly compatible with "play with it," whereas a park is not.

Therefore:

it → ball

Constraint 4: Coherence

Coherence considers the entire discourse rather than resolving each reference independently.

The paragraph introduces a natural sequence:

1. John and Mary go to the park.
2. John brings a ball.
3. Mary wants to play with the ball.
4. The dog chases John.
5. John, Mary, and the dog go home.

This produces consistent entity chains:

- **John:** John → He → him → they

- **Mary:** Mary → She → they
- **Ball:** ball → it
- **Dog:** dog → they

This interpretation gives a much more coherent overall story.

c) Constraint Graph / Table

Constraint Resolution Table

Variable	Domain	Gender/Number	Recency	Semantic Compatibility	Coherence	Final Link
He	John, Mary	John ✓, Mary ✗	John ✓	John ✓	John ✓	**John**
She	John, Mary	Mary ✓, John ✗	Mary ✓	Mary ✓	Mary ✓	**Mary**
it	ball, park	Both singular	ball ✓	ball ✓	ball ✓	**ball**
him	John, Mary, dog	John ✓, Mary ✗, dog ✓*	dog initially preferred	John ✓, dog less useful	John ✓	**John**
they	John, Mary, dog	plural group ✓	group ✓	group ✓	group ✓	**John + Mary + dog**
all	John, Mary, dog	plural group ✓	group ✓	group ✓	group ✓	**John + Mary + dog**

* “dog” can be referred to as “him” if the dog's gender is male, but the text does not establish that.

More importantly, the complete discourse is more coherent when **him = John**.

d) Final Coreference Chains

The final resolved chains are:

- **John → He → him → they**
- **Mary → She → they**
- **ball → it**
- **dog → they**
- **park** has no later referring expression.

Rewritten Paragraph

John and Mary went to the park. John brought a ball. Mary wanted to play with the ball. The dog chased John excitedly. Finally, John, Mary, and the dog all went home.

This rewritten version removes ambiguity while preserving the original meaning.

e) Priority Order of Constraints

A suitable priority order is:

1. Gender and number agreement → 2. Semantic compatibility → 3. Coherence → 4. Recency

1. Gender and Number Agreement

This is a strong grammatical constraint. For example, **"She" cannot refer to John** in the given context.

2. Semantic Compatibility

The candidate must make sense with the predicate. For example, "play with it" strongly supports **ball** rather than **park**.

3. Coherence

The final interpretation should produce a logical and consistent story. Coherence is particularly important when multiple candidates satisfy grammatical constraints.

4. Recency

Recency is useful as a preference, but it should not override stronger grammatical, semantic, and discourse constraints.

2. Constraint-Based Dialog Generation

Conversation History

User: “I have an important exam tomorrow but I’m not able to concentrate.”

Required Dialog Act: Advise + Encourage

a) Relevant Discourse Planning Steps

Before generating the response, the system should perform the following steps:

1. **Identify the user's problem:** The user has an important exam tomorrow and cannot concentrate.
2. **Identify the required dialog act:** The response must both **advise** and **encourage**.
3. **Maintain entity coherence:** Continue using the entities **exam**, **concentrate**, and **you**.
4. **Establish a discourse relation:** Use a **Cause-Effect** or **Elaboration** relationship.
5. **Select appropriate advice:** Recommend a short study/focus strategy such as taking a break and returning to study.
6. **Check length:** Ensure the final response contains only 2–3 sentences.
7. **Check politeness and logical consistency:** The advice should be supportive and realistic.

b) Three Possible Responses

Response 1

Take a short break to clear your mind, then return and focus on one small exam topic at a time. This can help you concentrate better, and you should feel confident that you can make good progress tonight.

Response 2

Since your exam is tomorrow, take a short break and then focus on the most important topics to improve your concentration. You have prepared for this, so stay confident and keep going.

Response 3

Your exam is important, so take a short break and use a quiet place to focus on your revision. When you reduce distractions, you can concentrate better and feel confident about your preparation.

c) Evaluation and Best Response

Criterion	Response 1	Response 2	Response 3
Advise + Encourage	✓	✓	✓
Entity coherence	✓	✓	✓
Uses exam/concentrate/focus	✓	✓	✓

Uses required keywords	✓	✓	✓
Cause-Effect/Elaboration	✓	✓	✓
2–3 sentences	✓	✓	✓
Logical consistency	✓	✓	✓
Positive/polite tone	✓	✓	✓
Overall coherence	Excellent	Excellent	Very good

Best Response: Response 1

Response 1 is the best choice because it provides a clear cause-effect progression:

short break → clearer mind → focused study → better concentration

It also maintains the entities **exam**, **concentrate**, and **you**, uses the required keywords **break, focus, and confident**, provides both advice and encouragement, and stays within the required two-sentence limit.

d) Effect of Violating Two Constraints

Violation 1: Entity Coherence

If the response stopped referring to the **exam** and **concentration** problem and suddenly discussed an unrelated topic, the dialog would become disconnected from the user's actual concern. This would reduce coherence and make the response appear irrelevant.

Violation 2: Response-Length Constraint

If the response contained many sentences instead of 2–3 sentences, it would fail the explicit generation constraint and become less efficient for a student who needs quick, practical advice. Even if the additional information were useful, the response would not satisfy the required specification.

3. Word Sense Disambiguation and Discourse Structure

Source Sentence

"The bank by the river flooded after the storm, but it was saved by quick action."

The word **"bank"** is ambiguous because it can mean:

1. A financial institution.
2. The land beside a river.

a) Resolve the Ambiguous Word "Bank"

The correct sense is:

bank = riverbank / land beside a river

Justification

The phrase:

"by the river"

strongly indicates that **bank** refers to the land beside the river.

The predicate:

"flooded after the storm"

also supports the riverbank interpretation because a riverbank can naturally flood.

A financial bank would normally be described with words such as **money, account, customers, loan, or transaction**, which are absent from the sentence.

Therefore:

bank → riverbank

Constraint-Based Decision

Constraint	Financial Bank	Riverbank
Context "by the river"	✗	✓
Can naturally flood	Weak	✓
Semantic compatibility	Weak	✓
Contextual coherence	✗	✓
Final sense	✗	✓

b) Predicate Logic Representation

Let:

- **b** = the bank/riverbank
- **r** = the river

- **s** = the storm
- **a** = quick action

Predicates:

- bank(b) = b is a bank/riverbank
- river(r) = r is a river
- location(b,r) = b is beside the river
- storm(s) = s is a storm
- flood(b) = the bank flooded
- after(flood(b), storm(s)) = the flooding occurred after the storm
- saved(b,a) = the bank was saved by quick action
- quick(a) = a was quick action

A suitable representation is:

∃b ∃r ∃s ∃a [

bank(b)

∧ river(r)

∧ location(b,r)

∧ storm(s)

∧ flood(b)

∧ after(flood(b), s)

∧ quick(a)

∧ saved(b,a)

]

The discourse relation between the two clauses can additionally be represented as:

Contrast(

flood(b) ∧ after(flood(b), s),

saved(b,a)

)

c) Target-Language Sentence / Simple English Paraphrase

Simple English Paraphrase

The riverbank flooded after the storm, but quick action saved it from further damage.

This preserves:

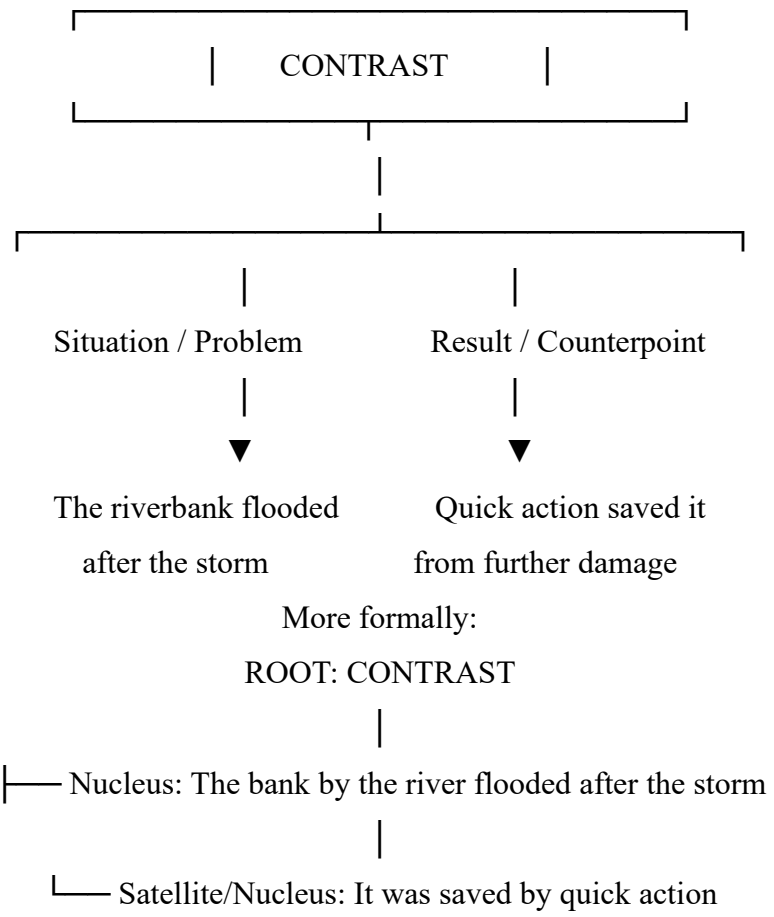
- **riverbank**
- **river**
- **storm**

- **flooding**
- **quick action**
- **saving**
- **contrast relation**

The ambiguous word **bank** has been explicitly replaced with **riverbank**, so the meaning is clear.

d) Simple RST-Style Discourse Tree

The sentence contains two main clauses connected by **"but."**



More formally:

ROOT: CONTRAST

├── Nucleus: The bank by the river flooded after the storm

└── Satellite/Nucleus: It was saved by quick action

The word **"but"** signals that the second clause contrasts with the expected consequence of the first clause: although the riverbank flooded, quick action prevented a worse outcome.

e) Advantages of Constraint-Based Approach Over Pure Statistical Machine Translation

A constraint-based approach has several important advantages.

1. Better Word Sense Disambiguation

The system explicitly uses the context **"by the river"** and **"flooded"** to select **riverbank** rather than financial institution.

2. Preservation of Logical Relations

The constraints ensure that:

bank → riverbank → flooded → saved

remain connected.

This reduces the possibility of losing important relationships during translation or generation.

3. Better Discourse Coherence

The **Contrast** relation introduced by “but” is explicitly preserved. Therefore, the output maintains the original discourse structure.

4. Reduced Information Loss

The approach explicitly requires preservation of important entities and relations such as:

- river
- riverbank
- storm
- flooding

Overall Constraint-Based Problem-Solving Strategy

The three problems demonstrate a common pipeline:

Input

↓

Identify Ambiguity / Variables

↓

Generate Candidate Domains

↓

Apply Hard Constraints

(Gender, Number, Grammar, Meaning)

↓

Apply Soft Constraints

(Recency, Coherence, Context)

↓

Rank Remaining Candidates

↓

Select Best Solution

↓

Check Global Consistency

↓

Generate Clear Output

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritizatio n	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solutio n Develo pment	Applicati on of Concept s	Justificati on	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director

Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____